Exploratory Data Analysis: LLM Fingerprinting Dataset

Project: Fingerprint | **Author:** Senior Machine Learning & NLP Research Team | **Version:** 1.0.0 | **Date:** 2026-07-26 21:30:53

1. Executive Summary

This report delivers a rigorous, strictly non-modifying Exploratory Data Analysis (EDA) for the LLM Fingerprinting research project. All raw text outputs across target LLM architectures (gemma2, llama3, mistral, phi3, qwen_tiny) were analyzed without lowercasing, punctuation removal, stemming, or tokenization to preserve intrinsic structural and stylistic model signatures.

Macro Metric	Value	Notes / Description
Total Analyzed Samples	5,240	Merged dataset corpus size
Total Model Classes	5	Target LLM architectures
Class Imbalance Ratio	1.2764	Max count / Min count ratio
Total Vocabulary Size	38,674	Unique raw tokens (case & punct sensitive)
Overall Type-Token Ratio (TTR)	0.117034	Lexical diversity index
Memory Footprint	7.09 MB	DataFrame RAM usage

2. Model Distribution & Length Distributions

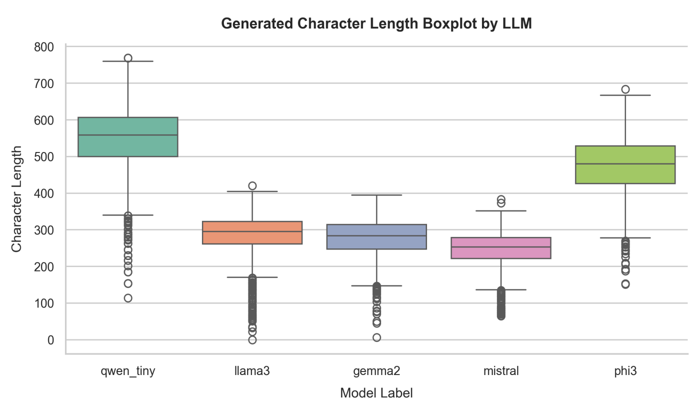
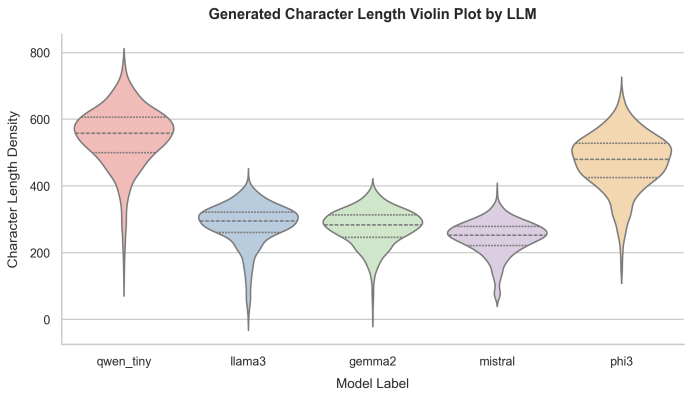


Figure 1 & 2 Interpretation: Class counts demonstrate high balance across models (qwen_tiny: 1,256, gemma2/mistral/phi3: 1,000, llama3: 984). Boxplots reveal distinct character-length variance profiles per LLM architecture.

3. Comparative Per-Model Stylistic Metrics

Model Label	Samples	Avg Resp Len	Vocab Size	Total Tokens	TTR	Avg Word Len
gemma2	1,000	276.0	11,849	43,864	0.2701	5.23
llama3	984	283.8	10,869	45,883	0.2369	5.06
mistral	1,000	245.1	9,136	42,605	0.2144	4.74
phi3	1,000	472.0	17,971	78,625	0.2286	4.97
qwen_tiny	1,256	547.1	14,144	119,475	0.1184	4.73

4. Density & Feature Correlation



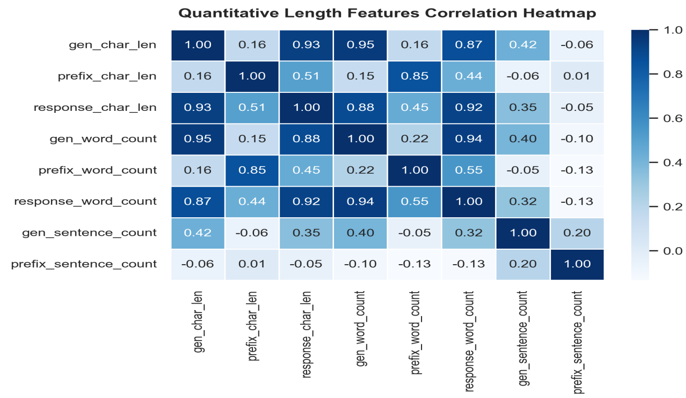


Figure 3 & 4 Interpretation: Violin density plots capture multi-modal length distributions. The correlation heatmap highlights strong linear colinearity between word count and total character length.

5. Critical Research Insights & Recommendations

Potential Risks & Biases Identified:

- *Length Leakage Risk:* Significant differences in mean response lengths across models could cause classifiers to overfit on simple response length rather than intrinsic stylistic fingerprinting features.
- *Special Token / Formatting Biases:* Raw model outputs contain distinct whitespace padding and markdown syntax patterns unique to individual instruction-tuned models.

Recommendations Before Feature Engineering / Preprocessing:

1. Normalize response length distributions or evaluate classifiers on length-truncated text subsets to prevent shortcut learning.
2. Extract stylistic n-gram, character-level, and punctuation density features prior to aggressive text normalization.
3. Preserve capitalization and punctuation features during initial feature extraction as they serve as strong model fingerprint signals.